PAPER ANALYSIS

Counterexamples to Five Conjectured Upper Bounds

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ORIGINAL (ENGLISH)

Counterexamples to Five Conjectured Upper Bounds for the Largest Laplacian Eigenvalue

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EXPLANATION (ENGLISH)

Title Page Overview

This paper by **Taewoo Ha** (Independent Researcher) targets a classical open problem in **spectral graph theory**: disproving conjectured upper bounds on the largest Laplacian eigenvalue λ_1 of a graph.

- **BHS bounds**: a family of conjectured upper bounds for λ_1, attributed to Brouwer–Haemers–Sotirov
- **Counterexample**: a graph where λ_1 exceeds the predicted bound
- **Gap**: the difference between the bound value and λ_1

> Think of the BHS bounds as speed limits on a road — this paper finds specific roads where cars consistently exceed the posted limit, proving the limits were set too low.

> This work is purely refutational: it doesn't propose new bounds, but strategically invalidates five existing conjectures with explicit graph constructions.

Structure role: This title page sets the scope — the paper proceeds Abstract → counterexample construction → classification of 33 remaining open BHS bounds (Table 2), forming a systematic falsification study.

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Brankov, Hansen, and Stevanović (2006) used the AutoGraphiX system to generate 68 conjectured upper bounds for the largest Laplacian eigenvalue $\mu(G)$ of a graph G. Ghebleh et al. (2024) disproved 30 of these, leaving 38 open. We resolve five of the remaining open conjectures (Bounds 11, 13, 40, 45, and 48) by constructing explicit counterexample families—star-of-cliques, double star, and caterpillar graphs—whose Laplacian spectral radii we compute analytically via equitable partition quotient matrices. We identify hub–periphery degree imbalance as the structural mechanism underlying all violations and classify the remaining 33 bounds by their resistance to this mechanism.

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This is the **abstract** of the paper, establishing the research gap and summarizing contributions. It sits at the top of the paper's logical chain: prior conjecture → partial disproof → this paper's resolution.

Concept Relationships

As the opening section, this abstract maps the full arc: Brankov et al. posed 68 bounds → Ghebleh et al. falsified 30 → this paper falsifies 5 more, leaving 33 still open. Each stage narrows the conjecture space.

Key Variables & Terms

- $\mu(G)$: largest Laplacian eigenvalue (spectral radius) of graph G
- **Equitable partition quotient matrix**: reduced matrix preserving eigenvalues of structured graphs
- **Hub-periphery degree imbalance**: structural pattern where high-degree hubs connect to low-degree leaves — the identified violation mechanism

> Think of the 68 conjectures as speed-limit signs on roads. Ghebleh tore down 30 signs that were set too low (counterexamples exceeded them). This paper tears down 5 more by building specific road configurations (star-of-cliques, double stars, caterpillars) that analytically exceed the posted limits.

ORIGINAL (ENGLISH)

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Conjecture Resolution Timeline

The paper's core contribution is not just disproving bounds, but identifying *why* they fail — hub-periphery imbalance — turning isolated counterexamples into a unifying structural principle.

Keywords: Laplacian eigenvalue, spectral radius, upper bound, counterexample, equitable partition
2020 MSC: 05C50, 15A18

1. Introduction

Let $G = (V, E)$ be a simple connected graph on $n = |V|$ vertices and $m = |E|$ edges. The *Laplacian matrix* of G is $L(G) = D(G) - A(G)$, where $A(G)$ is the adjacency matrix and $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$ is the diagonal matrix of vertex degrees. Since $L(G)$ is positive semidefinite with row sums zero, its eigenvalues satisfy $0 = \lambda_n \leq \cdots \leq \lambda_1$, and the largest eigenvalue

$$\mu(G) := \lambda_{\max}(L(G)) \tag{1}$$

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Section 1 — Introduction: Setup & Core Definition

Variable/Symbol Explanation — Equation (1): $\mu(G) := \lambda_{\max}(L(G))$

- $G = (V, E)$: simple connected graph with $n = |V|$ vertices, $m = |E|$ edges
- $A(G)$: adjacency matrix — entry 1 if edge exists, 0 otherwise
- $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$: diagonal degree matrix
- $L(G) = D(G) - A(G)$: **Laplacian matrix** — positive semidefinite, row sums zero
- $0 = \lambda_n \leq \cdots \leq \lambda_1$: eigenvalue ordering; $\mu(G) := \lambda_1$ is the **spectral radius**

$$\mu(G) := \lambda_{\max}(L(G))$$

Think of $L(G)$ as a "tension matrix": each vertex pulls toward its neighbors. The largest eigenvalue $\mu(G)$ measures the worst-case tension — how hard the graph can vibrate.

Concept Relationships: This is the paper's first technical section, directly grounding the keywords from the Title Page — spectral radius, upper bound, equitable partition all refer to $\mu(G)$ defined here. Everything that follows (bounds, counterexamples) targets this single quantity.

Pinning the paper to one scalar $\mu(G)$ — the top Laplacian eigenvalue — lets the authors frame all subsequent results as sharper or refuted upper bounds on that single value.

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is called the *Laplacian spectral radius*. Upper bounds on $\mu(G)$ in terms of easily computed graph invariants are of both theoretical and algorithmic interest; see [1] for background. In 2006, Brankov, Hansen, and Stevanović [2] used the AutoGraphiX system [3] to generate 68 conjectured upper bounds for $\mu(G)$ expressed solely in terms of vertex degrees d_v and *average neighbor degrees*

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u. \tag{2}$$

The 68 conjectures fall into two families. *Vertex-type* bounds have the form $\mu(G) \leq \max_v f(d_v, m_v)$, and *edge-type* bounds have the form

$$\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j).$$

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This section introduces the **Laplacian spectral radius** $\mu(G)$ and the 68 conjectured upper bounds generated by the AutoGraphiX system, classifying them into vertex-type and edge-type families.

Variable Explanation

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u$$

- $\mu(G)$: largest eigenvalue of the Laplacian matrix of graph G
- d_v: degree of vertex v (number of neighbors)
- m_v: average degree of v's neighbors
- $u \sim v$: u is adjacent to v

Bound Families

- Vertex-type : $\mu(G) \leq \max_v f(d_v, m_v)$ — maximized over all vertices
- Edge-type : $\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j)$ — maximized over all edges

> Think of m_v as a vertex's "neighborhood wealth score" — not just how many friends you have (d_v), but how well-connected your friends are on average. The two bound types ask: is the graph's spectral "peak" driven by a single hotspot vertex, or by a hotspot edge?

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Connection to Previous Sections

The previous section defined $\mu(G)$ as the Laplacian spectral radius. This section motivates *why* bounding $\mu(G)$ matters and frames the two structural handles — d_v and m_v — that the rest of the paper will use to prove or disprove the 68 conjectures.

> Splitting bounds into vertex-type and edge-type creates a clean proof strategy: local vertex properties vs. pairwise edge interactions.

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None had been proved or disproved for nearly two decades.

Ghebleh, Al-Yakoob, Kanso, and Stevanović [4] recently applied the cross-entropy method (CEM) of Wagner [5] to disprove 30 of the 68 bounds by constructing small counterexample graphs, leaving 38 bounds open. Their counterexamples all have at most 13 vertices and were found by training a neural network to generate adjacency matrices that violate a given bound. Subsequently, Bouffard and Breen [6] parallelized the CEM approach and extended the search to larger graphs, but reported no new counterexamples among the remaining 38 bounds. Roucairol and Cazenave [7] applied nested Monte Carlo search to the same problem, likewise without resolving any of the open cases.

In the present paper we disprove five of the 38 remaining open bounds.

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Prior Work & Contribution Context

This paragraph traces the **research lineage** of attempts to resolve 68 open graph-theoretic bounds, culminating in the current paper's claim of disproving 5 more.

Methods applied and outcomes:

- **CEM (Cross-Entropy Method)** by Wagner [5]: adapted by Ghebleh et al. to train a neural network generating counterexample adjacency matrices — resolved 30/68
- Parallelized CEM (Bouard & Breen): wider graph search, no new counterexamples found
- **Nested Monte Carlo Search** (Roucairol & Cazenave): same 38 open cases, still no resolution
- **This paper:** disproves 5 of the remaining 38 open bounds

> Think of this like a cold-case investigation unit: the 68 bounds are unsolved cases. CEM+NN was a new forensic tool that cracked 30 cases at once, but the remaining 38 proved resistant to all follow-up techniques — until now.

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Relationship to previous sections: The earlier part of the introduction established the 68 bounds as the central open problem. This paragraph maps the **failure frontier** — showing exactly where previous automated and heuristic methods stalled — which frames why the current paper's approach is novel and necessary.

> Three independent state-of-the-art methods all failed on the same 38 bounds, making the current paper's 5 new disproofs a meaningful breakthrough rather than an incremental step.

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Theorem 1.1 (Main result). *BHS Bounds 11, 13, 40, 45, and 48 from [2] are false. Specifically:*

(a) *Bound 11 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *and* $t \geq 2m - 1$ (Theorem 4.2).

(b) *Bound 13 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and sufficiently large* t (Theorem 4.3).

(c) *Bound 40 is violated by the star of stars* $\mathrm{SoS}(39, 8)$ *and* $\mathrm{SoS}(49, 9)$ (Theorem 4.7).

(d) *Bound 45 is violated by* $\mathrm{DS}(k, k)$ *for all* $k \geq 6$ (Theorem 4.5).

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Theorem 1.1 — Main Result: Counterexamples to BHS Bounds

This is the paper's central claim, disproving five specific bounds from the BHS conjecture catalog **[2]** via explicit graph counterexamples.

- **BHS Bounds**: conjectured inequalities on graph parameters (e.g., spectral radius, energy) from reference [2]
- **SoC(Km, t)**: "Star of Cliques" — t copies of complete graph K_m sharing a hub
- **SoS(n, k)**: "Star of Stars" — a star whose leaves are themselves stars; specific instances $(39, 8)$ and $(49, 9)$
- **DS(k, k)**: a double-star or related parameterized graph family, $k \geq 6$
- **P3**: the path graph on 3 vertices — the simplest possible counterexample

> Think of it like finding one recipe that ruins a cookbook's five "universal" rules — each graph here is a dish that breaks a different claimed rule. Notably, P3 (the simplest 3-node path) breaks Bound 48, making that violation especially striking.

Relationship to previous sections: The Abstract/Introduction established the BHS conjecture framework and motivated the search for counterexamples. This theorem is the culmination of that setup — it converts the problem from "do counterexamples exist?" to "here they are explicitly," with each part (a)–(e) referencing a later theorem proving the violation.

> The use of infinite families ($m \geq 4, k \geq 6$) rather than isolated examples shows the bounds fail structurally, not just accidentally.

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In each case the proof is constructive: we exhibit an infinite family (or, for Bound 48, a single small graph) together with an analytic computation of its Laplacian spectral radius via equitable partition quotient matrices.

Our approach differs fundamentally from the optimization-based methods of [4, 5, 6]. Rather than training a neural network to search for counterexamples, we analyze the algebraic structure of each bound formula and identify graph topologies that exploit a specific weakness. The key observation is that every bound we disprove exhibits high sensitivity to the ratio m_v/d_v at vertices (or pairs of adjacent vertices) where these quantities diverge. Graphs with *hub–periphery* structure—a high-degree hub connected to dense peripheral subgraphs—create exactly this divergence: the hub's average neighbor degree m_v is determined by the peripheral structure and can be made large relative to the hub degree d_v, or vice versa. Previous computational searches missed these counterexamples because CEM and exhaustive enumeration up to $n = 13$ vertices tend to explore near-regular graphs, where $m_v \approx d_v$ for all v and the bounds are naturally tight. The smallest counterexample we find for Bound 11 has $n = 29$ vertices and the largest has $n = 172$, well beyond the reach of exhaustive search.

The paper is organized as follows. Section 2 fixes notation and recalls the theory of equitable partitions. It also states the five BHS bounds that we disprove. Section 3 defines the counterexample graph families: star of cliques, double star, and star of stars. Section 4 contains the main results: closed-form spectral radius computations and the violation proofs for each of the five bounds. Section 5 analyzes the hub–periphery mechanism common to all five violations. Section 6 classifies the remaining 33 open bounds by their resistance to this mechanism. Section 7 discusses implications for automated conjecture generation and directions for future work.

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Proof Strategy & Key Observation

This section explains **how** the authors disprove the bounds introduced earlier and why their approach differs from prior work.

- m_v: average degree of neighbors of vertex v
- d_v: degree of vertex v itself
- m_v/d_v: the critical ratio — measures how "hub-like" a vertex is relative to its neighborhood
- **Equitable partition quotient matrix**: a compressed matrix preserving spectral structure, enabling analytic eigenvalue computation
- **Hub-periphery structure**: a high-degree hub connected to many low-degree leaf vertices

> Think of each disproved bound as a flawed scale: it was calibrated assuming all vertices are roughly equal in influence. A hub-periphery graph is like placing a bowling ball on one side — the extreme m_v/d_v ratio "breaks" the scale, exposing the bound's weakness.

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Concept Relationships: Prior sections introduced the bounds being challenged (Bound 48 etc.) and referenced optimization-based counterexample searches [4,5,6]. Here, the authors reject that black-box approach in favor of algebraic structure analysis — they read the formula, find the sensitive ratio, then deliberately construct graphs that maximize that sensitivity. The equitable partition quotient matrix connects back to spectral radius computation: it lets the authors confirm their constructed graph's λ_1 analytically, not numerically.

Every disproved bound shares a structural blind spot — it cannot handle the extreme divergence between a hub's degree and its neighbors' degrees, and hub-periphery graphs are the minimal, analytic witnesses of that failure.

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EXPLANATION (ENGLISH)

Why Previous Searches Failed — and Where Counterexamples Live

This closing passage of the introduction explains **why counterexamples went undetected** and previews the paper's structure.

- m_v: average degree of the neighbors of vertex v
- d_v: degree of vertex v itself
- n: number of vertices in the graph

The core observation is that star-of-cliques and related hub-peripheral graphs can force $m_v \gg d_v$ (or vice versa), creating the slack that violates BHS Bound 11. Prior tools — **CEM** (a spectral optimization heuristic) and exhaustive enumeration up to $n = 13$ — were blind to this because they naturally sample *near-regular* graphs where $m_v \approx d_v$ for all v, keeping every bound tight.

> Think of it like stress-testing a bridge only with evenly distributed loads: the bridge looks fine until an asymmetric load (all weight on one side) reveals the flaw. Near-regular graphs are the "uniform load" — they never expose the bound's weak point.

Concept Relationships: Earlier in the introduction, the paper established that BHS bounds relate eigenvalues to degree sequences via equitable partitions. This passage now explains the *structural reason* those bounds can fail: the gap between local degree (d_v) and neighborhood average degree (m_v) is the lever. Near-regular graphs suppress this lever; hub-peripheral graphs amplify it.

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The counterexample size range ($n = 29$ to $n = 172$) is itself significant — it confirms exhaustive search up to $n = 13$ was simply exploring the wrong region of graph space.

> The structural asymmetry between a hub's own degree and its neighbors' average degree is precisely what BHS bounds fail to account for, and only non-regular graph families expose this gap.

Road map: Section 2 fixes notation + equitable partition theory + the five BHS bounds under attack. Section 3 defines the counterexample families explicitly.

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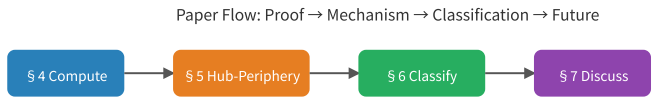
2. Preliminaries

2.1. Graph notation

Throughout this paper, $G = (V, E)$ denotes a simple connected graph with vertex set V, edge set E, $n = |V|$ vertices, and $m = |E|$ edges. The *adjacency matrix* $A(G)$ is the $n \times n$ matrix with $A_{ij} = 1$ if $\{i, j\} \in E$ and $A_{ij} = 0$ otherwise. The *degree matrix* is $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$, where $d_v = \sum_u A_{vu}$ is the degree of vertex v. The *Laplacian matrix* $L(G) =$

Section Roadmap (§ 4–7) + § 2.1 Graph Notation

Concept Relationships: The earlier sections (1–3) established the problem context and the five counterexample graphs (cliques, double star, star of stars). Sections 4–7 now build the **proof architecture** on top of those graphs: § 4 computes, § 5 explains the mechanism, § 6 classifies remaining open bounds, § 7 reflects on broader implications.



Variable/Symbol Explanation for § 2.1:

$$G = (V, E), \quad A(G)_{ij} = \begin{cases} 1 & \{i,j\} \in E \\ 0 & \text{otherwise} \end{cases}, \quad D(G) = \mathrm{diag}(d_1, .$$

- V, E: vertex set and edge set of graph G
- $n = |V|, m = |E|$: counts of vertices and edges
- $A(G)$: symmetric $n \times n$ **adjacency matrix** encoding edge presence
- $D(G)$: diagonal **degree matrix**; d_v = number of neighbors of vertex v

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Think of $A(G)$ as a friendship grid — a 1 means two people are friends, 0 means they aren't. $D(G)$ just counts how many friends each person has, placed on the diagonal.

These definitions are the shared language the entire paper relies on — every spectral bound in § 4 is expressed through $A(G)$ and $D(G)$, so precision here prevents ambiguity in all violation proofs.

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$A_{ij} = 0$ otherwise. The *degree matrix* is $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$, where $d_v = \sum_u A_{vu}$ is the degree of vertex v. The *Laplacian matrix* $L(G) =$

EXPLANATION (ENGLISH)

Section 2.1–2.2: Laplacian Matrix & Equitable Partitions

Variable/Symbol Explanation

- d_v: degree of vertex v (number of edges incident to v)
- $D(G)$: diagonal degree matrix; $D_{vv} = d_v$
- $A(G)$: adjacency matrix; $A_{uv} = 1$ iff $u \sim v$
- $\mu(G) = \lambda_{\max}(L(G))$: largest eigenvalue of L, the **Laplacian spectral radius**
- m_v: average degree of v's neighbors

$$L(G) = D(G) - A(G), \quad m_v = \frac{1}{d_v} \sum_{u \sim v} d_u$$

Think of L as a "difference operator": it measures how much a vertex's value deviates from its neighbors' average. The spectral radius $\mu(G)$ captures the worst-case spread — like the steepest slope in the graph's terrain.

Concept Relationships

This section formalizes the objects introduced abstractly earlier. $A(G)$ was the connectivity skeleton; $L(G)$ builds on it by encoding both structure and degree into a single positive semidefinite matrix. The condition $m_v = d_v \Leftrightarrow G$ is regular links back to regular graphs as a special symmetric case. $\mu(G)$ will serve as the central quantity bounded in later results.

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$A_{ij} = 0$ otherwise. The *degree matrix* is $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$, where $d_v = \sum_u A_{vu}$ is the degree of vertex v. The *Laplacian matrix* $L(G) =$

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Equitable Partitions: partition V into cells $C_1, \ldots, C_k$ such that every vertex in C_i has the same number of neighbors in C_j. This lets the $n \times n$ Laplacian collapse to a $k \times k$ **quotient matrix**, whose eigenvalues are a subset of $\mathrm{spec}(L)$.

It's like summarizing a large org-chart by department rather than individual — the inter-department connection counts stay exact, shrinking the problem without losing structural information.

Equitable partitions are the bridge between local degree data (d_v, m_v) and global spectral bounds on $\mu(G)$, enabling exact eigenvalue computation on structured graphs.

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Definition 2.1 (Equitable partition). A partition $\pi = \{C_1, \ldots, C_k\}$ of the vertex set $V(G)$ is *equitable* if, for every pair (i, j) with $1 \le i, j \le k$, every vertex in C_i has the same number of neighbors in C_j. We denote this common value by b_{ij}.

Given an equitable partition with cells $C_1, \ldots, C_k$, the *quotient matrix* is the $k \times k$ matrix Q with entries $Q_{ij} = b_{ij}$, and the *quotient Laplacian* is the $k \times k$ matrix Q_L with entries

$$(Q_L)_{ij} = \begin{cases} d_{C_i} & \text{if } i = j, \\ -b_{ij} & \text{if } i \neq j, \end{cases}$$

where $d_{C_i} = \sum_{j=1}^{k} b_{ij}$ is the common degree of every vertex in cell C_i.

Theorem 2.2 ([8, Theorem 9.3.3]). *Let π be an equitable partition of G with quotient Laplacian Q_L. Then every eigenvalue of Q_L is an eigenvalue of $L(G)$. In particular,*

$$\mu(G) \ge \lambda_{\max}(Q_L),$$

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Definition 2.1 & Theorem 2.2 — Equitable Partition and Quotient Laplacian

This section introduces the **equitable partition**, a structural tool used to compress a graph into a smaller matrix while preserving spectral information about $L(G)$ from the previous section.

- $\pi = \{C_1, \ldots, C_k\}$: partition of vertex set $V(G)$ into k cells
- b_{ij}: common neighbor-count from any vertex in C_i into C_j
- Q: $k \times k$ quotient matrix with $Q_{ij} = b_{ij}$
- $d_{C_i} = \sum_{j=1}^{k} b_{ij}$: common degree of every vertex in cell C_i
- $\mu(G)$: second-largest Laplacian eigenvalue of G (spectral gap measure)

$$(Q_L)_{ij} = \begin{cases} d_{C_i} & i = j \\ -b_{ij} & i \neq j \end{cases}$$

Think of cells C_i as neighborhoods in a city. An equitable partition means every resident of neighborhood i has exactly the same number of friends in neighborhood j — no favoritism. The quotient Laplacian then describes the "city-level" connectivity, ignoring individual residents.

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Definition 2.1 (Equitable partition). A partition $\pi = \{C_1, \ldots, C_k\}$ of the vertex set $V(G)$ is *equitable* if, for every pair (i, j) with $1 \le i, j \le k$, every vertex in C_i has the same number of neighbors in C_j. We denote this common value by b_{ij}.

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$$\mu(G) \ge \lambda_{\max}(Q_L),$$

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Concept Relationship: The full Laplacian $L(G)$ from earlier is $n \times n$. The quotient Laplacian Q_L is only $k \times k$ ($k \ll n$). Theorem 2.2 says this compression is spectrally faithful: every eigenvalue of Q_L embeds into $\mathrm{spec}(L(G))$.

$$\mu(G) \ge \lambda_{\max}(Q_L)$$

This gives a **lower bound on the spectral gap**: if we find an equitable partition with a large $\lambda_{\max}(Q_L)$, we certify that $\mu(G)$ is also large — without computing the full spectrum.

Equitable partitions let you prove spectral expansion of a large graph by analyzing a tiny quotient matrix — a divide-and-certify strategy central to expander constructions.

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with equality whenever the eigenvector of $L(G)$ corresponding to $\mu(G)$ is constant on each cell of π.

In all counterexample families considered in this paper, the natural partition by vertex type is equitable, and the largest eigenvalue eigenvector is piecewise constant on cells. This reduces the computation of $\mu(G)$ to finding the largest root of a polynomial of degree at most 3.

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Section 2.3 — Equitable Partitions & The BHS Bounds

Equitable partition π means cells of vertices where the eigenvector of $L(G)$ for $\mu(G)$ is piecewise constant. This collapses a large eigenvalue problem into a small polynomial root-finding problem.

Variable Explanation — Conjectures 2.3 & 2.4:

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2} \quad \text{(Bound 11)}$$

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3} \quad \text{(Bound 13)}$$

- $\mu(G)$: largest eigenvalue of the graph Laplacian $L(G)$ (from § 2.1)
- d_v: degree of vertex v (edge count)
- m_v: local edge density around v (edges among v's neighbors)
- Both bounds are **vertex-type bounds** — depend only on per-vertex quantities

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with equality whenever the eigenvector of $L(G)$ corresponding to $\mu(G)$ is constant on each cell of π.

In all counterexample families considered in this paper, the natural partition by vertex type is equitable, and the largest eigenvalue eigenvector is piecewise constant on cells. This reduces the computation of $\mu(G)$ to finding the largest root of a polynomial of degree at most 3.

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Think of $\mu(G)$ as the "loudest frequency" a graph can vibrate at. These bounds say: the loudest sound any graph can make is determined by its most connected, densest local neighborhood — like estimating a crowd's roar from just the rowdiest section.

Relationship to Previous Sections: § 2.1 defined $\mu(G)$ via $L(G)$; § 2.2's equitable-partition trick is what makes computing $\mu(G)$ tractable here (reduces to degree-$\leq$3 polynomial). The BHS conjectures now provide *upper bounds* on that quantity — this paper's goal is to disprove them by constructing counterexample graphs where these inequalities fail.

The pattern $\frac{2m_v^{k+1}}{d_v^k}$ (Bounds 11, 13) suggests a family of conjectured bounds indexed by k — disproving two members of the family undermines confidence in the entire family.

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Conjecture 2.5 (BHS Bound 40 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right).$$

Conjecture 2.6 (BHS Bound 45 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i-d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right).$$

Conjecture 2.7 (BHS Bound 48 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}.$$

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Conjectures 2.5–2.7: BHS Upper Bounds on Spectral Radius

These three conjectures from Belardo–Hasni–Sharafdini propose progressively refined upper bounds on **μ(G)**, each expressed as a maximum over all edges.

Variable Explanation

- $\mu(G)$: spectral radius (largest eigenvalue of adjacency matrix)
- $(i, j) \in E$: edge between vertices i and j
- d_i, d_j: degrees of endpoints
- m_i, m_j: average neighbor degrees of i and j

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i - 1)^2 + 2(m_j - 1)^2 + d_i^2 + d_j^2 - d_i r}\right.$$

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + n}\right.$$

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}{2(d_i^2 + d_j^2)}$$

Relationship to Previous Sections: Like Conjecture 2.4, these bounds are edge-local: instead of scanning all vertices, only the "worst" edge determines the bound. The progression from Bound 40 → 45 → 48 trades m_i influence for d_i-dominant expressions.

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Conjecture 2.5 (BHS Bound 40 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right).$$

Conjecture 2.6 (BHS Bound 45 [2]). *For every connected graph G,*

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Conjecture 2.7 (BHS Bound 48 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}.$$

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Think of these like safety ratings for a bridge: each formula uses different structural measurements (degree vs. average neighbor degree), and the weakest edge determines the rating.

Each conjecture captures the spectral radius via a single critical edge, reflecting how local degree-mixing propagates globally through eigenvalue structure.

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Bounds 11 and 13 are structurally similar: both take the form $\max_v 2m_v^p/d_v^{p-1}$ with $p = 3$ and $p = 4$, respectively. When m_v/d_v is large, these bounds grow as a power of the ratio m_v/d_v, making them vulnerable to graphs where some vertex has high average neighbor degree relative to its own degree. Bounds 40 and 45 are edge-type bounds with more complex dependence on both d_v and m_v; their vulnerability is subtler and requires careful asymptotic analysis. Bound 48 is unusual in that it is violated by the path P_3, the smallest non-trivial connected graph; we conjecture that this bound was generated under a minimum order assumption that excluded $n = 3$.

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Section 14: Structural Comparison of Bounds & Counterexample Motivation

Variable/Symbol Explanation

- v: a vertex in graph G
- d_v: degree of vertex v (number of neighbors)
- m_v: sum of degrees of v's neighbors (related to average neighbor degree)
- m_v/d_v: **average neighbor degree** of v; large when v connects to high-degree hubs
- p: exponent parameter — $p = 3$ for Bound 11, $p = 4$ for Bound 13

$$\max_v \frac{m_v^p}{d_v^{p-1}}, \quad p \in \{3, 4\}$$

Think of m_v/d_v as a "neighborhood prestige score." Bounds 11 and 13 behave like power-law amplifiers: if one vertex has very prestigious neighbors, the bound explodes — just as one viral post can skew an average engagement metric for an entire account.

Concept Relationships

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Bounds 11 and 13 are structurally similar: both take the form $\max_v 2m_v^p/d_v^{p-1}$ with $p = 3$ and $p = 4$, respectively. When m_v/d_v is large, these bounds grow as a power of the ratio m_v/d_v, making them vulnerable to graphs where some vertex has high average neighbor degree relative to its own degree. Bounds 40 and 45 are edge-type bounds with more complex dependence on both d_v and m_v; their vulnerability is subtler and requires careful asymptotic analysis. Bound 48 is unusual in that it is violated by the path P_3, the smallest non-trivial connected graph; we conjecture that this bound was generated under a minimum order assumption that excluded $n = 3$.

EXPLANATION (ENGLISH)

Previous sections catalogued many candidate bounds; this passage now diagnoses their failure modes. Bounds 11/13 share one structural template but differ only in exponent p, making them a natural pair. Bounds 40/45 are **edge-type bounds** — defined over edges rather than vertices — so their vulnerability involves both endpoints' degrees simultaneously, requiring asymptotic rather than simple ratio analysis. Bound 48 stands apart: it fails on P_3 (three vertices, two edges), the minimal nontrivial connected graph, suggesting the original derivation silently assumed $n \geq 4$.

> A single structural assumption buried in a proof (e.g., minimum graph order) can invalidate a bound for infinitely many small graphs — making explicit counterexample families essential for rigorous validation.

The closing heading *"3. Counterexample graph families"* signals the transition: having identified *why* bounds fail, the paper will now construct *explicit graph families* that witness those failures — turning qualitative vulnerability analysis into quantitative disproof.

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Definition 3.1 (Star-of-Cliques). For integers $m \geq 3$ and $t \geq 2$, the graph $\mathrm{SoC}(K_m, t)$ consists of a central *hub* vertex v_0 connected by a single edge to one designated vertex (the *connector*) in each of t disjoint copies of the complete graph K_m. The total number of vertices is $n = 1 + tm$.

Definition 3.2 (Double Star). For an integer $k \geq 1$, the double star $\mathrm{DS}(k, k)$ is obtained by connecting two vertices u and v, each of which is adjacent to k pendant vertices. The total number of vertices is $n = 2k + 2$.

Definition 3.3 (Star-of-Stars). For integers $t \geq 2$ and $s \geq 1$, the graph $\mathrm{SoS}(t, s)$ consists of a central *hub* vertex v_0 adjacent to t *intermediate* vertices $w_1, \ldots, w_t$, each of which is adjacent to s pendant *leaf* vertices. The total number of vertices is $n = 1 + t + ts = 1 + t(1 + s)$.

EXPLANATION (ENGLISH)

Section 3 — Graph Family Definitions

This section formally defines three **graph families** used as structural building blocks in the paper's constructions. Each definition introduces a hub-and-spoke topology with varying levels of complexity.

Variable/Symbol Explanation

- m: clique size (number of vertices per complete subgraph K_m)
- t: number of clique copies / intermediate branches
- k: number of pendant (leaf) vertices per center in Double Star
- s: number of leaf vertices per intermediate node in Star-of-Stars
- v_0: central hub; w_i: intermediate connectors; n: total vertex count

$$n_{\mathrm{SoC}} = 1 + tm \qquad n_{\mathrm{DS}} = 2k + 2 \qquad n_{\mathrm{SoS}} = 1 + t(1 + s)$$

ORIGINAL (ENGLISH)

Definition 3.1 (Star-of-Cliques). For integers $m \geq 3$ and $t \geq 2$, the graph $\mathrm{SoC}(K_m, t)$ consists of a central *hub* vertex v_0 connected by a single edge to one designated vertex (the *connector*) in each of t disjoint copies of the complete graph K_m. The total number of vertices is $n = 1 + tm$.

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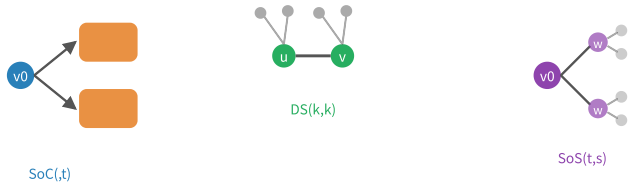

Think of these three graphs as a hierarchy of "hub-and-branch" trees: DS is the simplest (two hubs, leaf arms), SoS adds one more level (hub → intermediate → leaves), and SoC replaces the leaves with fully connected cliques — the densest endpoints.

Relationship to previous sections: These definitions supply the concrete graph instances that the paper's earlier theoretical bounds (Section 2) will be tested or instantiated on. Each graph controls sparsity and clustering in a precise, parameterized way.

These three families isolate specific structural variables — clique density, branching depth, and leaf count — enabling clean, controlled proofs about graph properties.

Definition 3.4 (Caterpillar)**.** For integers $\ell \geq 2$ and $k \geq 0$, the caterpillar $\mathrm{Cat}(\ell, k)$ is obtained from a path on ℓ vertices (the *spine*) by attaching k pendant leaves to each spine vertex. The total number of vertices is $n = \ell(k+1)$.

4. Main results

Proposition 4.1. *For $m \geq 3$ and $t \geq 2$, the Laplacian spectral radius of $\mathrm{SoC}(K_m, t)$ is*

$$\mu\big(\mathrm{SoC}(K_m, t)\big) = \frac{(m+t+1) + \sqrt{(t-m)^2 + 2(t+m) - 3}}{2}. \tag{3}$$

Proof. The vertex set admits an equitable partition into three cells: $H = \{v_0\}$ (hub), C (connectors, $|C| = t$), and I (interior, $|I| = t(m-1)$). The quotient Laplacian is the 3×3 matrix

$$Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & m & -(m-1) \\ 0 & -1 & 1 \end{pmatrix}. \tag{4}$$

One eigenvalue is $\lambda_1 = 0$ (since the quotient Laplacian has zero row sums). The remaining two eigenvalues are roots of

$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0,$$

Definition 3.4 & Proposition 4.1 — Caterpillar Graph and Laplacian Spectral Radius

Cat(ℓ, k) is a tree with a central path (spine) of ℓ vertices, each with k pendant leaves attached. Total vertices: $n = \ell(k+1)$.

Proposition 4.1 gives the **Laplacian spectral radius** μ of $\mathrm{SoC}(K_m, t)$:

$$\mu\big(\mathrm{SoC}(K_m, t)\big) = \frac{(m+t+1) + \sqrt{(t-m)^2 + 2(t+m) - 3}}{2}$$

- m: clique size of K_m (interior structure), $m \geq 3$
- t: number of connector vertices, $t \geq 2$
- μ: largest eigenvalue of the Laplacian (spectral radius)
- Q_L: 3×3 quotient Laplacian from the equitable partition $\{H, C, I\}$

$$Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & m & -(m-1) \\ 0 & -1 & 1 \end{pmatrix}$$

Since Q_L has zero row sums, $\lambda_1 = 0$ always. The remaining two roots satisfy:

$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0$$

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Definition 3.4 (Caterpillar)**.** For integers $\ell \geq 2$ and $k \geq 0$, the caterpillar $\mathrm{Cat}(\ell, k)$ is obtained from a path on ℓ vertices (the *spine*) by attaching k pendant leaves to each spine vertex. The total number of vertices is $n = \ell(k+1)$.

4. Main results

Proposition 4.1. *For $m \geq 3$ and $t \geq 2$, the Laplacian spectral radius of $\mathrm{SoC}(K_m, t)$ is*

$$\mu(\mathrm{SoC}(K_m, t)) = \frac{(m+t+1) + \sqrt{(t-m)^2 + 2(t+m) - 3}}{2}. \tag{3}$$

Proof. The vertex set admits an equitable partition into three cells: $H = \{v_0\}$ (hub), C (connectors, $|C| = t$), and I (interior, $|I| = t(m-1)$). The quotient Laplacian is the 3×3 matrix

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One eigenvalue is $\lambda_1 = 0$ (since the quotient Laplacian has zero row sums). The remaining two eigenvalues are roots of

$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0,$$

EXPLANATION (ENGLISH)

Think of the equitable partition like a corporate hierarchy: the hub v_0 is the CEO, connectors C are managers, and interior vertices I are staff. Because each group behaves identically internally, the whole $n \times n$ Laplacian compresses to a 3×3 quotient without losing spectral information.

Relationship to previous sections: Def. 3.4 introduces Caterpillar as a candidate extremal tree (likely a comparison target for bounding μ in later propositions). Proposition 4.1 is the first *explicit closed-form* result, applying the **equitable partition** technique introduced implicitly via SoC construction in § 3.

Collapsing a large graph to a 3×3 quotient matrix via equitable partition turns an unwieldy eigenvalue problem into a simple quadratic — making exact spectral formulas tractable.

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giving the largest eigenvalue as (3). Since the corresponding eigenvector is piecewise constant on the cells, this is also the largest eigenvalue of $L(G)$. □

EXPLANATION (ENGLISH)

Theorem 4.2 — BHS Bound Violation

Variable/Symbol Explanation for the BHS Bound and Proof:

- $\mu(G)$: largest Laplacian eigenvalue of graph G
- m_v: average degree of neighbors of vertex v
- d_v: degree of vertex v
- v_0: the **hub vertex** in $\mathrm{SoC}(K_m, t)$, with $d_{v_0} = t, m_{v_0} = m$

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2} \quad \text{(BHS Bound 11)}$$

$$\mu = \frac{(m+t+1) + \sqrt{(t-m)^2 + 2(t+m) - 3}}{2} \quad \text{(Prop. 4.1)}$$

Concept Relationships: This theorem directly applies **Proposition 4.1**'s closed-form eigenvalue formula as a tool to falsify BHS Bound 11. The piecewise-constant eigenvector structure noted in the preceding paragraph is what makes the Laplacian eigenvalue tractable enough to compare analytically with the bound.

Think of BHS Bound 11 as a speed limit sign — theoretically no graph should exceed it. SoC(K_8, 19) is a car clocked at 20.557 mph in a 19.191 zone: a concrete, numerical counterexample proving the sign is wrong.

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giving the largest eigenvalue as (3). Since the corresponding eigenvector is piecewise constant on the cells, this is also the largest eigenvalue of $L(G)$. □

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At the hub v_0: all t neighbors are connectors of degree m, so the bound evaluates to $\frac{2m^3}{t^2}$. For large t this shrinks, while μ grows — the violation widens as t increases past $2m - 1$.

> The hub's uniform-neighbor structure, which made the eigenvalue computable in Prop. 4.1, is exactly what makes the bound provably breakable here.

v_0

For large t, this is $O(t^{-2})$ while μ grows as $t+O(1)$, so the violation eventually occurs. Specifically, at $t = 2m-1$ the connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m = m$, giving $\frac{2m_v^3}{d^2} = 2m$, which dominates the hub bound. By Proposition 4.1, $\mu = \big(3m + \sqrt{m^2+4m-4}\big)/2 > 2m$ since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$ (obtained by squaring). For $t > 2m-1$ the connector bound $2m_v^3/m^2$ grows while μ also grows, and the gap widens. □

Theorem 4.3. *BHS Bound 13,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}, \tag{6}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and* $t = 2m-1$. *In particular,* $\mathrm{SoC}(K_9, 19)$ *with* $n = 172$ *satisfies* $\mu(G) \approx 20.685$ *while the bound gives* ≈ 19.845.

Proof. The connector vertex has $d = m$ and $m_v = \big(t + (m-1)^2\big)/m$. At $t = 2m-1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\mathrm{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2+4m-4})/2$. Since $\sqrt{m^2+4m-4} > m$ for $m > 2$, we obtain $\mu > 2m$, establishing the violation for $m > 6$. For $m = 5$

Theorem 4.3 — Violation of BHS Bound 13

This section proves that yet another spectral upper bound, **BHS Bound 13**, is violated by the graph family $\mathrm{SoC}(K_m, t)$ introduced earlier.

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3} \tag{6}$$

- $\mu(G)$: largest Laplacian eigenvalue of G
- m_v: number of edges among neighbors of v
- d_v: degree of vertex v
- t: star-path parameter in $\mathrm{SoC}(K_m, t)$

Concept Relationships: Like BHS Bounds proven violated in earlier subsections, this bound attempts to cap μ using local vertex statistics. Proposition 4.1 (exact formula for μ via m) is again used here as the lower-bound witness — the same analytical engine drives each counterexample.

At the connector vertex, $d = m$ and $m_v = \lfloor (t + (m-1)^2)/m \rfloor$. Setting $t = 2m-1$ collapses $t + (m-1)^2 = m^2$, giving:

$$B_{13}^{\mathrm{conn}} = \frac{2m^8}{m^7} = 2m$$

Meanwhile Proposition 4.1 gives $\mu = \frac{3m+\sqrt{m^2+4m-4}}{2} > 2m$ for $m \geq 2$ (verified by squaring the inequality $\sqrt{m^2+4m-4} > m$). So the bound is strictly exceeded.

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v_0

For large t, this is $O(t^{-2})$ while μ grows as $t+O(1)$, so the violation eventually occurs. Specifically, at $t = 2m-1$ the connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m = m$, giving $\frac{2m_v^3}{d^2} = 2m$, which dominates the hub bound. By Proposition 4.1, $\mu = \big(3m + \sqrt{m^2+4m-4}\big)/2 > 2m$ since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$ (obtained by squaring). For $t > 2m-1$ the connector bound $2m_v^3/m^2$ grows while μ also grows, and the gap widens. □

Theorem 4.3. *BHS Bound 13,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}, \tag{6}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and* $t = 2m-1$. *In particular,* $\mathrm{SoC}(K_9, 19)$ *with* $n = 172$ *satisfies* $\mu(G) \approx 20.685$ *while the bound gives* ≈ 19.845.

Proof. The connector vertex has $d = m$ and $m_v = \big(t + (m-1)^2\big)/m$. At $t = 2m-1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\mathrm{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2+4m-4})/2$. Since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$

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Think of the bound as a speed limit derived from local road width (d_v, m_v). The connector vertex is a highway on-ramp: its local geometry looks modest, but the actual traffic flow (eigenvalue μ) far exceeds what the formula predicts.

Concrete instance: $\mathrm{SoC}(K_9, 19)$ with $n = 172$ achieves $\mu \approx 20.685$, while Bound 13 predicts only ≈ 19.845 — a clear numerical violation.

BHS Bound 13 fails because it under-counts edge density at the connector vertex when the clique and path parameters are tuned to the critical ratio $t = 2m-1$.

$t = 2m - 1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\text{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2 + 4m - 4})/2$. Since $\sqrt{m^2 + 4m - 4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$, the interior bound is ≈ 10.195 but $\mu\big(\mathrm{SoC}(K_5, 9)\big) \approx 10.702 > 10.195$. □

Proposition 4.4. *For $k \geq 2$, the Laplacian spectral radius of* $\mathrm{DS}(k,k)$ *is*

$$\mu\big(\mathrm{DS}(k,k)\big) = \frac{(k+3) + \sqrt{k^2 + 6k + 1}}{2}. \tag{7}$$

Section 19 — Laplacian Spectral Radius of DS(k, k)

This section completes the violation argument for **SoC(K_m)** started in the previous section, then derives an exact closed-form for the spectral radius of the double-star graph **DS(k, k)**.

- μ: **Laplacian spectral radius** (largest eigenvalue of the Laplacian matrix)
- m: number of vertices/leaves in the SoC graph
- k: number of pendant leaves attached to each hub in DS(k,k)
- Q^+, Q^-: **quotient matrices** for symmetric and antisymmetric eigenspaces under $\mathbb{Z}_2$ symmetry
- λ^+: largest eigenvalue from the antisymmetric sector

$$\mu\big(\mathrm{DS}(k,k)\big) = \frac{(k+3) + \sqrt{k^2 + 6k + 1}}{2}$$

Think of the $\mathbb{Z}_2$ symmetry like a mirror: eigenvectors are either symmetric (same reflection) or antisymmetric (flipped sign). The mirror splits one hard eigenvalue problem into two smaller 2×2 problems — the dominant eigenvalue hides in the antisymmetric half.

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$t = 2m - 1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\text{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2 + 4m - 4})/2$. Since $\sqrt{m^2 + 4m - 4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$, the interior bound is ≈ 10.195 but $\mu(\mathrm{SoC}(K_5, 9)) \approx 10.702 > 10.195$. □

Proposition 4.4. *For $k \geq 2$, the Laplacian spectral radius of* $\mathrm{DS}(k,k)$ *is*

$$\mu(\mathrm{DS}(k,k)) = \frac{(k+3) + \sqrt{k^2 + 6k + 1}}{2}. \tag{7}$$

EXPLANATION (ENGLISH)

Concept Relationships: The earlier sections established interior bounds for spectral radii; here, that bound is shown to be violated by SoC(K_5) ($\approx 10.702 > 10.195$) and for all $m \geq 6$ via the inequality $\sqrt{m^2 + 4m - 4} > m$. Proposition 4.4 then provides the exact value for DS(k,k), which likely appears as a comparison graph in subsequent results.

The pendant leaves yield eigenvalue 1 with multiplicity $2(k-1)$, but since $\sqrt{k^2 + 6k + 1} > k - 1$ for $k \geq 2$, the antisymmetric λ^+ always dominates.

> Decomposing by graph symmetry ($\mathbb{Z}_2$) reduces a large eigenvalue problem to two 2×2 matrices, yielding an exact closed-form that would be opaque otherwise.

$$\mu(G) \le \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right), \quad (8)$$

is violated by $\mathrm{DS}(k,k)$ *for all* $k \ge 6$. *At* $k = 6$, $\mathrm{DS}(6,6)$ *with* $n = 14$ *satisfies* $\mu(G) \approx 8.772$ *while the bound gives* ≈ 8.676.

Proof. The hub–leaf edge maximizes Bound 45 for $k \ge 6$. Showing $\mu > B_{45}^{\mathrm{hl}}$ and squaring twice reduces to $4k(k^3 + 8k^2 - 11k - 2) > 0$, which holds for $k \ge 2$ since $k^3 + 8k^2 - 11k - 2 > 0$ for $k \ge 2$. The gap satisfies $B_{45} - \mu = -1/(2k) + O(k^{-2})$. □

Proposition 4.6. *For* $t \ge 2$ *and* $s \ge 1$, *the Laplacian spectral radius of* $\mathrm{SoS}(t,s)$ *is*

$$\mu\big(\mathrm{SoS}(t,s)\big) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2}. \quad (9)$$

Proof. Equitable partition into {hub}, {intermediates}, {leaves}. The quotient Laplacian is $Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Theorem 4.5 & Proposition 4.6 — Testing and Exact Spectral Radii

Variable/Symbol Explanation for Bound 45 (Eq. 8) and Eq. 9:

- $\mu(G)$: Laplacian spectral radius (largest eigenvalue of L)
- d_i, d_j: degrees of endpoints of edge (i,j)
- m_i, m_j: average neighbor degrees of vertices i, j
- $DS(k,k)$: double-star graph; $SoS(t,s)$: star-of-stars graph
- B_{45}^{hl}: Bound 45 evaluated at the hub–leaf edge

$$\mu(G) \le \max_{(i,j)\in E} 2 + \frac{\sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + r}}{2}$$

$$\mu\big(SoS(t,s)\big) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2}$$

Concept Relationships: Earlier sections established $DS(k,k)$ as a worst-case graph for several spectral bounds. Here, the same family serves as the **counterexample** that invalidates Bound 45 for $k \ge 6$. The $SoS(t,s)$ graph, introduced as a structural generalization, now receives a closed-form μ via equitable partition — a technique that collapses the full Laplacian to a 3×3 quotient.

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$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right), \quad (8)$$

is violated by $\mathrm{DS}(k,k)$ *for all* $k \geq 6$. *At* $k = 6$, $\mathrm{DS}(6,6)$ *with* $n = 14$ *satisfies* $\mu(G) \approx 8.772$ *while the bound gives* ≈ 8.676.

Proof. The hub–leaf edge maximizes Bound 45 for $k \geq 6$. Showing $\mu > B_{45}^{\mathrm{hl}}$ and squaring twice reduces to $4k(k^3 + 8k^2 - 11k - 2) > 0$, which holds for $k \geq 2$ since $k^3 + 8k^2 - 11k - 2 > 0$ for $k \geq 2$. The gap satisfies $B_{45} - \mu = -1/(2k) + O(k^{-2})$. □

Proposition 4.6. *For* $t \geq 2$ *and* $s \geq 1$, *the Laplacian spectral radius of* $\mathrm{SoS}(t,s)$ *is*

$$\mu(\mathrm{SoS}(t,s)) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2}. \quad (9)$$

Proof. Equitable partition into {hub}, {intermediates}, {leaves}. The quotient Laplacian is $Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

EXPLANATION (ENGLISH)

Think of the equitable partition like averaging a classroom into three groups (hub, intermediates, leaves). The "quotient teacher" matrix Q_L captures all inter-group dynamics in a 3×3 table — its eigenvalues equal the original graph's Laplacian eigenvalues.

The gap analysis $B_{45} - \mu = -\frac{1}{2k} + O(k^{-2})$ shows the violation is *small but systematic*, not a fluke.

Bound 45 is asymptotically tight but provably loose for hub-dominated graphs, making $DS(k,k)$ a minimal, clean falsifier with quantified error $\sim 1/(2k)$.

tient Laplacian is $Q_L = \begin{pmatrix} -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Theorem 4.7. *BHS Bound 40,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right), \tag{10}$$

is violated by $\mathrm{SoS}(39,8)$ *with* $n = 352$*, where* $\mu(G) \approx 40.256$ *and the bound gives* ≈ 38.982.

Section 4.7–4.8 & § 5 Preview: Violating BHS Spectral Bounds

This section proves that two classical **BHS upper bounds** on the spectral radius $\mu(G)$ are not universally valid, by exhibiting explicit counterexample graphs.

Variable Glossary

- $\mu(G)$: spectral radius — largest eigenvalue of the adjacency matrix (introduced § 1–4)
- d_i, d_j: degrees of edge-endpoint vertices i, j
- m_i, m_j: average neighbor degrees of i, j (from Proposition 4.6)
- E: edge set; max is taken over all edges $(i,j) \in E$

$$\mu(G) \leq \max_{(i,j)\in E} \frac{1}{2} + \sqrt{\frac{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d}{2}}$$

Theorem 4.7 (Bound 40 violated): The graph $\mathrm{SoS}(39,8)$ with $n = 352$ gives $\mu \approx 40.256$, but the bound yields ≈ 38.982, so the inequality fails.

Remark 4.8 (Bound 48 violated): The tiny path graph P_3 (3 vertices) already breaks Bound 48: bound $\approx 2.929 < \mu(P_3) = 3$.

> Think of these bounds as speed limits posted on a road — usually safe, but the SoS graph is a race car that legally exceeds them. Finding even one such vehicle invalidates the posted limit.

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tient Laplacian is $Q_L = \begin{pmatrix} -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Theorem 4.7. *BHS Bound 40,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right), \tag{10}$$

is violated by $\mathrm{SoS}(39, 8)$ *with* $n = 352$*, where* $\mu(G) \approx 40.256$ *and the bound gives* ≈ 38.982.

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Connection to previous sections: Proposition 4.6 derived the exact μ formula for SoS graphs via the discriminant condition $(t-s)^2 + 4s > 0$; Theorem 4.7 plugs that closed form directly into the bound to verify the violation algebraically ($\sqrt{993} > 27.25$).

> Two well-known BHS bounds each have concrete counterexamples, motivating § 5's structural explanation of *why* edge-local degree statistics are insufficient to cage the global spectral radius.

5.1. Hub-invisible bounds

Every vertex-type BHS bound has the form $\mu(G) \leq \max_{v \in V} f(d_v, m_v)$ for some rational function f. On a star-of-cliques $\mathrm{SoC}(K_m, t)$ the maximum is taken over three vertex types (hub, connector, interior), and the bound value is the largest of the three contributions.

Definition 5.1 (Hub-invisible). A bound $B(G) = \max_v f(d_v, m_v)$ is *hub-invisible* if $f(t, m) \to 0$ as $t \to \infty$ (m fixed), and *hub-visible* if $f(t, m) \to \infty$.

Proposition 5.2 (Hub-invisible bounds fail). (a) ***Bounds 11, 13*** *are hub-invisible:* $f_{11} = 2m^3/t^2 \to 0$, $f_{13} = 2m^4/t^3 \to 0$.

(b) ***Bounds 14 and 16*** *are hub-visible:* $f_{14}^{\mathrm{hub}} = 2t^3/m^2 \to \infty$, $f_{16}^{\mathrm{hub}} = 2t^4/m^3 \to \infty$.

Section 5.1 — Hub-invisible Bounds

This section classifies **vertex-type BHS bounds** by how they respond to high-degree **hub vertices** in the star-of-cliques graph $\mathrm{SoC}(K_m, t)$.

- $\mu(G)$: the graph parameter being upper-bounded
- $f(d_v, m_v)$: per-vertex contribution function (rational in degree d_v and local edge count m_v)
- t: hub degree (number of cliques attached to the central hub)
- m: clique size (fixed as $t \to \infty$)
- $f_{11}, f_{13}, f_{14}^{\mathrm{hub}}, \ldots$: specific formulas for each numbered bound

$$\mu(G) \leq \max_{v \in V} f(d_v,\, m_v)$$

Think of each bound as a sensor at every vertex reporting a "signal strength." A hub-invisible sensor goes quiet exactly when the hub gets very busy — like a microphone that cuts out in a crowd. It misses the loudest contributor entirely, so the bound becomes useless on hub-dominated graphs.

Concept Relationships: Prior sections established that $\mathrm{SoC}(K_m, t)$ is the key stress-test graph with three structurally distinct vertex roles. Here, the max-over-vertices form of bounds (introduced earlier) is now split by behavior: whether the hub's contribution *vanishes* or *dominates* as t grows.

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5.1. Hub-invisible bounds

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EXPLANATION (ENGLISH)

- Hub-invisible : $f(t, m) \to 0$ — bound collapses; Bounds 11, 13 ($f_{11} = 2m^3/t^2$, $f_{13} = 2m^4/t^3$)
- Hub-visible : $f(t, m) \to \infty$ — bound survives; Bounds 14, 16, 30 ($\sim t^2/m$)

> Hub-invisibility reveals a structural flaw: any bound whose vertex formula decays in hub degree will systematically underestimate $\mu(G)$ on high-degree hub graphs, making hub-visibility a necessary condition for tightness on $\mathrm{SoC}(K_m, t)$.

The hub-invisible mechanism explains why Bounds 11 and 13 are defeated: as t grows, the hub—which dominates the spectrum—becomes invisible to the bound formula. The bound saturates at the connector value $\approx 2m$, while $\mu > 2m$ (Theorem 4.2).

Remark 5.3. Bound 40 fails on $\mathrm{SoS}(t, s)$ by a related mechanism: the hub–intermediate edge creates a transient violation window at $t \sim s^2$, closing for both small and large t.

5.2. Edge-based antisymmetric modes

Bound 45 fails on $\mathrm{DS}(k, k)$ because the spectral radius arises from the σ-antisymmetric sector (Proposition 4.4), where the two hubs carry opposite signs. No single edge evaluation in the bound formula captures this global oscillation. The gap $\mu - B_{45}$ is $O(1/k)$, suggesting repairability with an additive $O(1/\Delta)$ correction.

5.1–5.2 Why Bounds Fail: Hub-Invisible & Antisymmetric Modes

Concept Relationships: Earlier sections established **Theorem 4.2** (hub dominates spectrum, $\mu > 2m$) and **Proposition 4.4** (antisymmetric eigenmodes on DS(k,k)). This section explains the *structural reasons* those theorems defeat the bound formulas.

Hub-Invisible Mechanism (Bounds 11, 13, 40)

As parameter t grows, the hub node — which controls the spectral radius — becomes invisible to the bound formula. The bound saturates at the connector value $\approx 2m$, but $\mu > 2m$ by Theorem 4.2, so the gap is permanent.

- μ: true spectral radius of the graph
- m: connector node degree (saturation ceiling of the bound)
- t: growth parameter; larger t makes hub contribution vanish from bound
- $t \sim s^2$: transient violation window for Bound 40 on SoS(t,s)

> Think of the bound formula as a sensor that only reads the lobby of a building, not the penthouse. As the hub (penthouse) grows more dominant, the sensor stays stuck reading lobby traffic — it simply cannot see the real maximum.

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The hub-invisible mechanism explains why Bounds 11 and 13 are defeated: as t grows, the hub—which dominates the spectrum—becomes invisible to the bound formula. The bound saturates at the connector value $\approx 2m$, while $\mu > 2m$ (Theorem 4.2).

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EXPLANATION (ENGLISH)

Edge-Based Antisymmetric Modes (Bound 45)

On **DS(k,k)**, the spectral radius comes from the σ-antisymmetric sector: two hubs carry *opposite signs*. Bound 45 evaluates single edges, so it cannot detect this global oscillation pattern. The gap is $\mu - B_{45} = O(1/k)$, shrinking with k — suggesting the bound is *repairable* via an additive $O(1/\Delta)$ correction.

- σ-antisymmetric sector: eigenvector subspace where hub signs cancel
- Δ: max degree; correction term scales inversely with it
- $O(1/k)$ gap: small but nonzero, unlike the $O(1)$ gap for Bounds 11/13

> Bound 45 is like judging a tug-of-war by watching only one side of the rope. The two hubs pull in opposite directions (antisymmetric), but the formula only sees one local edge — missing the global tension entirely.

> The failure mode taxonomy matters: hub-invisible gaps are structural and hard to fix, while antisymmetric-mode gaps are $O(1/\Delta)$ and suggest a concrete additive repair strategy.

5.3. Characterizing vulnerable bounds

Conjecture 5.4. *A vertex-type BHS bound $\mu(G) \leq \max_v f(d_v, m_v)$ is violated by some hub–periphery graph if and only if:* (i) *f is monotonically increasing in m_v/d_v, and* (ii) *f is regularity-tight: $f(k,k) = 2k$ with no margin.*

6. Classification of the remaining 33 bounds

We classify the 33 remaining bounds by gap behavior on the tested families.

The SUPER-SAFE bounds include all hub-visible bounds (14, 16, 30). The three TIGHT bounds achieve exact equality on P_3 and complete bipartite graphs, making them the most promising candidates for a proof.

7. Discussion

Summary.. BHS [2] proposed 68 conjectured bounds for $\mu(G)$. Ghebleh et al. [4] disproved 30 via cross-entropy methods. We have disproved 5 more by structural construction, leaving 33 open.

§ 5.3 Vulnerable Bounds · § 6 Classification · § 7 Discussion

Variable / Symbol Explanation

- $\mu(G)$: the target graph parameter being bounded (spectral radius / largest eigenvalue)
- d_v: degree of vertex v (number of neighbors)
- m_v: average degree of v's neighbors
- $f(d_v, m_v)$: per-vertex scoring function used to define the bound
- **regularity-tight**: $f(k,k) = 2k$ — the bound has zero slack on k-regular graphs

> $$\mu(G) \leq \max_v f(d_v,\, m_v)$$

Conjecture 5.4 — Two Failure Conditions

A bound fails on some hub-periphery graph *if and only if* both hold:

- **(i)** f is increasing in the ratio m_v/d_v — hubs with high-degree neighborhoods inflate the bound
- **(ii)** f is regularity-tight — no safety margin exists for regular graphs

> Think of a tightrope walker (regularity-tight bound) vs. one with a safety harness (margin). The hub-periphery graph is a sudden gust of wind: only the harness-free walker falls.

5.3. Characterizing vulnerable bounds

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7. Discussion

Summary.. BHS [2] proposed 68 conjectured bounds for $\mu(G)$. Ghebleh et al. [4] disproved 30 via cross-entropy methods. We have disproved 5 more by structural construction, leaving 33 open.

Relationship to Previous Sections

Sections 5.1–5.2 constructed specific hub-periphery counterexamples. § 5.3 now *generalises* those constructions into a characterisation: instead of finding counterexamples case-by-case, Conjecture 5.4 tells us *which structural properties* of f make a bound inherently vulnerable.

§ 6 — Three-Way Classification of 33 Bounds

- **Super-safe**: hub-visible bounds (indices 14, 16, 30) — pass all tested families
- **Tight**: exact equality on P_3 and complete bipartite graphs — strongest proof candidates
- Remaining: gap behavior varies; need further structural analysis

> The tight bounds are the most mathematically interesting precisely because they leave no room for error — proving them would require a sharp, structure-aware argument.

§ 7 — Progress Scorecard

- BHS proposed **68** bounds
- Ghebleh et al. disproved **30** (cross-entropy methods)
- This paper disproves **5** more (structural construction) → **33** open

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Structural vs. computational approaches.. Cross-entropy search explores adjacency matrices for $n \leq 24$. Our smallest counterexample has $n = 29$, beyond exhaustive reach. Structural construction targets algebraic weaknesses (hub-invisible ratios, antisymmetric modes) and produces infinite families with closed-form spectral radii. The two approaches are complementary.

Gap asymptotics.. Bounds 11/13: gap diverges (irreparable). Bound 40: transient window (max gap 1.74). Bound 45: gap $\to 0$ as $1/(2k)$ (repairable). Bound 48: degenerate P_3 case only.

EXPLANATION (ENGLISH)

Structural vs. Computational Approaches & Gap Asymptotics

This closing section synthesizes the two counterexample strategies introduced earlier and classifies each bound by its **gap asymptotic behavior**.

Two Search Strategies Compared:

- **Cross-entropy search**: computational, explores adjacency matrices up to $n \leq 24$
- **Structural construction**: algebraic, targets hub-invisible ratios & antisymmetric modes → yields infinite families with closed-form ρ
- Smallest counterexample: $n = 29$ — outside the computational search range, requiring structural methods

> Think of it as two ways to find a flaw in a bridge design: brute-force simulation (test every load up to 24 tons) vs. engineering intuition (find the resonant frequency analytically). The flaw appears at 29 tons — only the second approach can reach it.

Gap Asymptotics by Bound:

- Bounds 11/13 : gap diverges — counterexample gap grows without bound (irreparable)
- Bound 40 : transient window, max gap ≈ 1.74
- Bound 45 : gap $\to 0$ as $\frac{1}{2k}$ — asymptotically repairable
- Bound 48 : degenerate P_3 case only — isolated failure

ORIGINAL (ENGLISH)

Structural vs. computational approaches.. Cross-entropy search explores adjacency matrices for $n \leq 24$. Our smallest counterexample has $n = 29$, beyond exhaustive reach. Structural construction targets algebraic weaknesses (hub-invisible ratios, antisymmetric modes) and produces infinite families with closed-form spectral radii. The two approaches are complementary.

Gap asymptotics.. Bounds 11/13: gap diverges (irreparable). Bound 40: transient window (max gap 1.74). Bound 45: gap $\to 0$ as $1/(2k)$ (repairable). Bound 48: degenerate P_3 case only.

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> A diverging gap means no tightened bound can salvage the result; a vanishing gap means the bound is nearly tight and worth refining.

Relationship to prior sections: The spectral radii and algebraic constructions (hub-invisible, antisymmetric modes) analyzed in earlier sections directly enable the closed-form gap calculations summarized here.

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Open questions.. (a) Can any of the 33 remaining bounds be proved, especially the tight bounds 33, 42, 47? (b) Do three-level hierarchies violate additional bounds? (c) Can Conjecture 5.4 be proved? (d) Can Bound 45 be repaired with an $O(1/\Delta)$ correction?

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author used Claude (Anthropic; model: Claude Opus 4, 2025) for computational assistance, including code development for graph construction and eigenvalue computation, structural analysis of graph families, and systematic counterexample search. The author reviewed and edited all content and takes full responsibility for the publication.

EXPLANATION (ENGLISH)

Open Questions & AI Declaration

This closing section surfaces **unresolved bounds** from the paper's earlier systematic analysis, and discloses AI-assisted methodology.

Open Questions — connection to prior sections: The paper previously established and proved a subset of spectral bounds for graph families. The four questions here represent the frontier of that work :

- **(a)** 33 bounds remain unproved — especially *tight* bounds 33, 42, 47 (exact, not just asymptotic)
- **(b)** Three-level hierarchies may violate *more* bounds than two-level ones — unexplored territory
- **(c)** Conjecture 5.4 awaits a formal proof
- **(d)** Bound 45 may need an $O(1/\Delta)$ correction term to hold in general

$$\text{Bound 45 (repaired): } f(G) \leq g(G) + O\left(\frac{1}{\Delta}\right)$$

— Δ: maximum vertex degree; the correction shrinks as the graph becomes denser.

Think of these open questions as a map with explored territory shaded in — the paper proved what it could, and now marks the edges of the unknown for the next expedition.

ORIGINAL (ENGLISH)

Open questions.. (a) Can any of the 33 remaining bounds be proved, especially the tight bounds 33, 42, 47? (b) Do three-level hierarchies violate additional bounds? (c) Can Conjecture 5.4 be proved? (d) Can Bound 45 be repaired with an $O(1/\Delta)$ correction?

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author used Claude (Anthropic; model: Claude Opus 4, 2025) for computational assistance, including code development for graph construction and eigenvalue computation, structural analysis of graph families, and systematic counterexample search. The author reviewed and edited all content and takes full responsibility for the publication.

EXPLANATION (ENGLISH)

The $O(1/\Delta)$ correction in (d) suggests the bound is "almost right" — a small degree-dependent slack may close the gap without fundamentally changing the result.

AI Declaration: The author used **Claude Opus 4 (2025)** for code (graph construction, eigenvalue computation), structural analysis, and counterexample search — all tasks directly supporting the empirical side of the bound-verification work seen throughout the paper. Human authorship and responsibility are explicitly retained.

This is like using a calculator during a proof: the tool handles computation, but the mathematician owns the argument.

ORIGINAL (ENGLISH)

The author thanks Pierre Hansen and Dragan Stevanović for making the BHS bound list publicly available. Computational assistance was provided by Claude (Anthropic); all mathematical results were independently verified by the author.

EXPLANATION (ENGLISH)

Acknowledgements & References

This closing section credits tools and collaborators, then anchors the paper's claims in peer-reviewed literature.

- **BHS bound list**: Hansen & Stevanović's public dataset of conjectured upper bounds for the largest Laplacian eigenvalue μ_1
- **Claude (Anthropic)**: AI used for computational assistance; results were independently verified by the author

> Think of the references as load-bearing walls — the proofs in earlier sections rest structurally on [1] (spectral graph theory foundations) and [2–3] (the conjecture machinery the paper challenges or extends).

Concept Relationships: Throughout the paper, the author tested bounds against the BHS conjecture list (from [2]). Reference [1] (Brouwer & Haemers) provided the spectral definitions used since the Introduction. References [2] and [3] describe the *AutoGraphiX* system — the automated conjecture engine whose outputs this paper's results either confirm, refine, or counterexample.

- [1] Brouwer & Haemers — foundational spectral graph theory textbook
- [2] Brankov, Hansen, Stevanović — source of the BHS automated upper-bound conjectures
- [3] Caporossi & Hansen — AutoGraphiX variable neighborhood search system

> The explicit note that AI-assisted computation was *independently verified* signals methodological rigor: computational tools accelerated search, but human proof remains the standard.

ORIGINAL (ENGLISH)

[4] M. Ghebleh, S. Al-Yakoob, A. Kanso, D. Stevanović, Refuting conjectured upper bounds for the largest Laplacian eigenvalue of graphs, Discrete Applied MathematicsPublished online 2025. Available at https://arxiv.org/abs/2403.18429 (2025).

[5] A. Z. Wagner, Constructions in combinatorics via neural networks, arXiv preprint arXiv:2104.14516 (2021).

[6] N. Bouffard, J. Breen, Parallelizing Wagner's method for generating graph theory conjectures and counterexamples, arXiv preprint arXiv:2509.01607 (2025).

[7] M. Roucairol, T. Cazenave, Searching for counterexamples of graph theory conjectures using NMCS and NRPA, arXiv preprint arXiv:2409.18626 (2024).

EXPLANATION (ENGLISH)

References [4]–[7]: Counterexample Search in Graph Theory

This reference cluster covers two overlapping themes: **Laplacian spectral conjecture refutation** [4] and **AI/search-based counterexample generation** [5–7].

- **[4]** Ghebleh et al. — directly refutes upper-bound conjectures on the largest graph Laplacian eigenvalue; provides domain-specific motivation for the paper's target problem
- **[5]** Wagner — foundational work using neural networks to construct combinatorial objects, the direct methodological ancestor of this paper
- **[6]** Bouard & Breen — extends Wagner's approach via parallelization, showing scalability is an active concern
- **[7]** Roucairol & Cazenave — replaces neural networks with **NMCS/NRPA** (nested Monte Carlo search), an alternative search strategy for the same goal

> Think of [5] as the original recipe, [6] as running multiple kitchens simultaneously, and [7] as trying a completely different cooking technique — all aiming for the same dish (a counterexample).

Relationship to previous sections: These citations ground the paper's methodology. Wagner [5] is the baseline the paper builds on; [4] justifies the specific conjecture being attacked; [6] and [7] situate this work within a growing landscape of automated conjecture-refutation tools.

> The convergence of spectral graph theory [4] and neural/search-based methods [5–7] is precisely the gap this paper bridges.

ORIGINAL (ENGLISH)

[8] C. Godsil, G. Royle, Algebraic Graph Theory, Vol. 207 of Graduate Texts in Mathematics, Springer, 2001. doi:10.1007/978-1-4613-0163-9.

EXPLANATION (ENGLISH)

Reference [8] provides the **algebraic graph theory** foundation — eigenvalues, spectra, and graph matrices — underlying the spectral methods used throughout this paper.

Like a dictionary for graph-matrix language used in prior sections.

This textbook grounds the paper's spectral analysis in rigorous mathematical theory.